

Chapter 2

Quadratic Equations

Only One Option Correct Type Questions

1. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is [IIT-JEE-2007 (Paper-1)]

(A) $\frac{2}{9}(p - q)(2q - p)$

(B) $\frac{2}{9}(q - p)(2p - q)$

(C) $\frac{2}{9}(q - 2p)(2q - p)$

(D) $\frac{2}{9}(2p - p)(2q - p)$

2. Let p and q be real numbers such that $p \neq 0, p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [IIT-JEE-2010 (Paper-1)]

(A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$

(B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$

(D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

3. Let β and α be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ [IIT-JEE-2011 (Paper-1)]

(A) 1

(B) 2

(C) 3

(D) 4

4. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is [IIT-JEE-2011 (Paper-2)]

(A) $-\sqrt{2}$

(B) $-i\sqrt{3}$

(C) $i\sqrt{5}$

(D) $\sqrt{2}$

5. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has [JEE (Adv)-2014 (Paper-2)]
- (A) Only purely imaginary roots
 (B) All real roots
 (C) Two real and two purely imaginary roots
 (D) Neither real nor purely imaginary roots
6. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is [JEE (Adv)-2020 (Paper-1)]
- (A) 0 (B) 8000
 (C) 8080 (D) 16000

One or More Option(s) Correct Type Questions

7. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? [JEE (Adv)-2015 (Paper-2)]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$
 (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

8. Let α and β be the roots of $x^2 - x - 1 = 0$, with $\alpha > \beta$. For all positive integers n , define

$$a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \quad n \geq 1,$$

$$b_1 = 1 \text{ and } b_n = a_{n-1} + a_{n+1}, \quad n \geq 2$$

Then which of the following options is/are correct?

[JEE (Adv)-2019 (Paper-1)]

- (A) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$
 (B) $b_n = \alpha^n + \beta^n$ for all $n \geq 1$
 (C) $a_1 + a_2 + a_3 + \dots + a_n = a_{n+2} - 1$ for all $n \geq 1$
 (D) $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 9 and 10

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

[JEE (Adv)-2017 (Paper-2)]

Choose the correct answer :

9. If $a_4 = 28$, then $p + 2q =$

(A) 12

(B) 21

(C) 14

(D) 7

10. $a_{12} =$

(A) $a_{11} + 2a_{10}$

(B) $a_{11} + a_{10}$

(C) $a_{11} - a_{10}$

(D) $2a_{11} + a_{10}$

Matrix-Match / Matrix Based Type Questions

11. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Match the expressions/statements in **Column-I** with expressions/statements in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS. [IIT-JEE-2007 (Paper-2)]

Column-I

(A) If $-1 < x < 1$, then $f(x)$ satisfies

(B) If $1 < x < 2$, then $f(x)$ satisfies

(C) If $3 < x < 5$, then $f(x)$ satisfies

(D) If $x > 5$, then $f(x)$ satisfies

Column-II

(p) $0 < f(x) < 1$

(q) $f(x) < 0$

(r) $f(x) > 0$

(s) $f(x) < 1$

Integer / Numerical Value Type Questions

12. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is

[IIT-JEE-2011 (Paper-2)]

13. For $x \in \mathbb{R}$, the number of real roots of the equation

[JEE (Adv)-2021 (Paper-1)]

$3x^2 - 4|x^2 - 1| + x - 1 = 0$ is _____.

